EXOPLANETS · DISCOVERY

A temperate sub-Neptune transiting the nearby M dwarf MTS-0412

A. Okafor, L. Brandt, S. Ishikawa and the Meridian Transit Survey team

Abstract. *We report a transiting planet candidate, MTS-0412 b, found in 118 nights of photometry from the Meridian southern array. Nine transits of depth 2,400 ± 60 ppm recur every 11.82 days. For the host's radius of 0.41 solar radii the depth implies a planet of 2.2 Earth radii, which receives roughly 1.3 times the flux the Earth receives from the Sun. A second, shallower signal at 4.07 days remains below our detection threshold.*

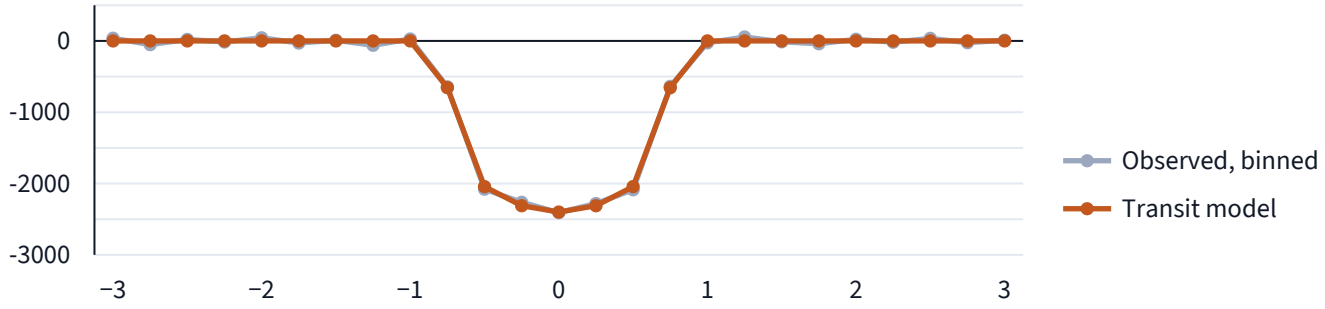


Figure 1. Nine transits of MTS-0412 b, folded on the 11.82-day period and binned to 15 minutes. The flux change, in parts per million, is plotted against hours from mid-transit; the model is a limb-darkened transit of depth 2,400 ppm.

1 Introduction

Small planets around M dwarfs offer the most favourable transit geometry for atmospheric study: the star is small, so a planet of a few Earth radii blocks a measurable fraction of its light. The Meridian survey monitors 3,200 such stars south of declination −30° with a cadence of two minutes.

2 Observations

MTS-0412 was observed on 118 nights between February and September 2026. Each night's frames were calibrated, aligned and measured by differential aperture photometry against eight comparison stars of similar colour, using version 3.4 of the survey pipeline. The combined light curve has a scatter of 410 ppm per two-minute point.

3 Results

A box least-squares search of the detrended light curve recovers a period of 11.8204 ± 0.0007 days. The transit lasts 1.7 hours and is flat-bottomed, which excludes a grazing eclipsing binary. Table 1 lists every candidate above a signal-to-noise ratio of seven.

Star	Period	Depth	Radius	Status
MTS-0412 b	11.820	2,400	2.2	Candidate
MTS-1187 b	3.514	5,900	3.1	Confirmed
MTS-0903 b	7.266	1,150	1.4	Candidate
MTS-2240 b	1.902	14,300	9.6	Binary
MTS-0412 c	4.070	610	1.1	Marginal

Table 1. Candidates above a signal-to-noise ratio of seven: period in days, depth in ppm, radius in Earth radii.

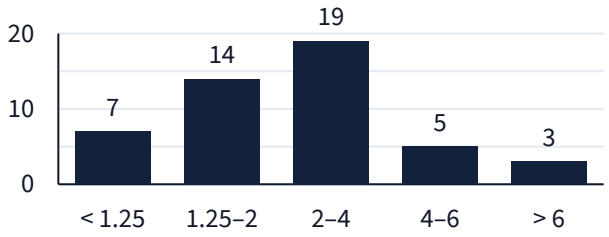


Figure 2. All 48 survey candidates by radius, in Earth radii.

4 Discussion

MTS-0412 b lies above the radius valley and is most likely a sub-Neptune with a hydrogen envelope. Radial velocities in 2027 should settle its mass.